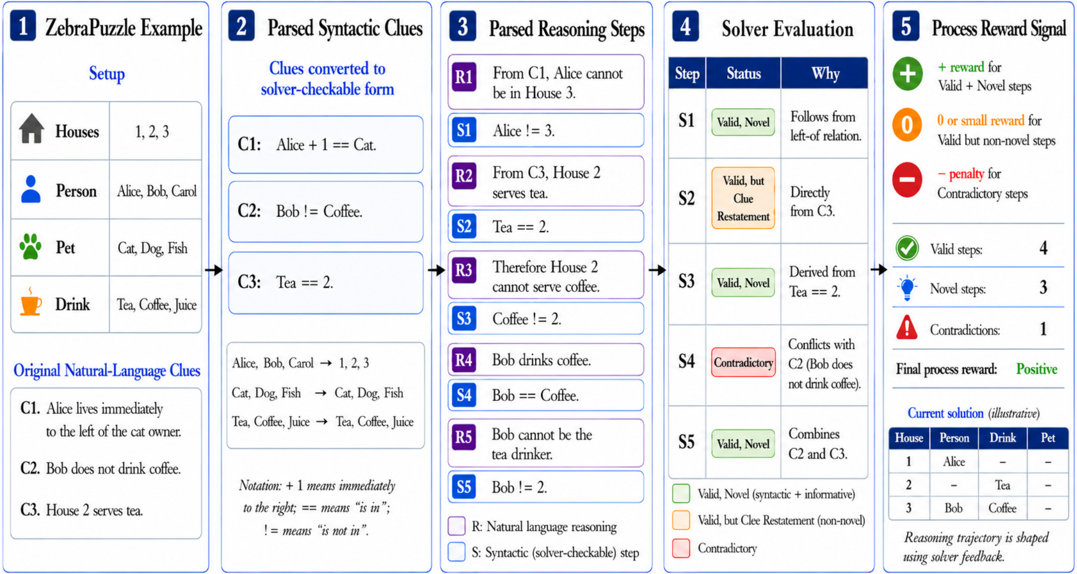
1 ZebraPuzzle Example
Setup
Houses 1, 2, 3
Person Alice, Bob, Carol
Pet Cat, Dog, Fish
Drink Tea, Coffee, Juice
Original Natural-Language Clues
C1. Alice lives immediately to the left of the cat owner.
C2. Bob does not drink coffee.
C3. House 2 serves tea.
2 Parsed Syntactic Clues
Clues converted to solver-checkable form
C1: Alice + 1 == Cat.
C2: Bob != Coffee.
C3: Tea == 2.
Alice, Bob, Carol → 1, 2, 3
Cat, Dog, Fish → Cat, Dog, Fish
Tea, Coffee, Juice → Tea, Coffee, Juice
Notation: + 1 means immediately to the right; == means "is in"; != means "is not in".
3 Parsed Reasoning Steps
R1 From C1, Alice cannot be in House 3.
S1 Alice != 3.
R2 From C3, House 2 serves tea.
S2 Tea == 2.
R3 Therefore House 2 cannot serve coffee.
S3 Coffee != 2.
R4 Bob drinks coffee.
S4 Bob == Coffee.
R5 Bob cannot be the tea drinker.
S5 Bob != 2.
R: Natural language reasoning
S: Syntactic (solver-checkable) step
4 Solver Evaluation
Step | Status | Why
S1 | Valid, Novel | Follows from left-of relation.
S2 | Valid, but Clue Restatement | Directly from C3.
S3 | Valid, Novel | Derived from Tea == 2.
S4 | Contradictory | Conflicts with C2 (Bob does not drink coffee).
S5 | Valid, Novel | Combines C2 and C3.
Valid, Novel (syntactic + informative)
Valid, but Clee Restatement (non-novel)
Contradictory
5 Process Reward Signal
+ reward for Valid + Novel steps
0 or small reward for Valid but non-novel steps
− penalty for Contradictory steps
Valid steps: 4
Novel steps: 3
Contradictions: 1
Final process reward: Positive
Current solution (illustrative)
House | Person | Drink | Pet
1 | Alice | – | –
2 | – | Tea | –
3 | Bob | Coffee | –
Reasoning trajectory is shaped using solver feedback.